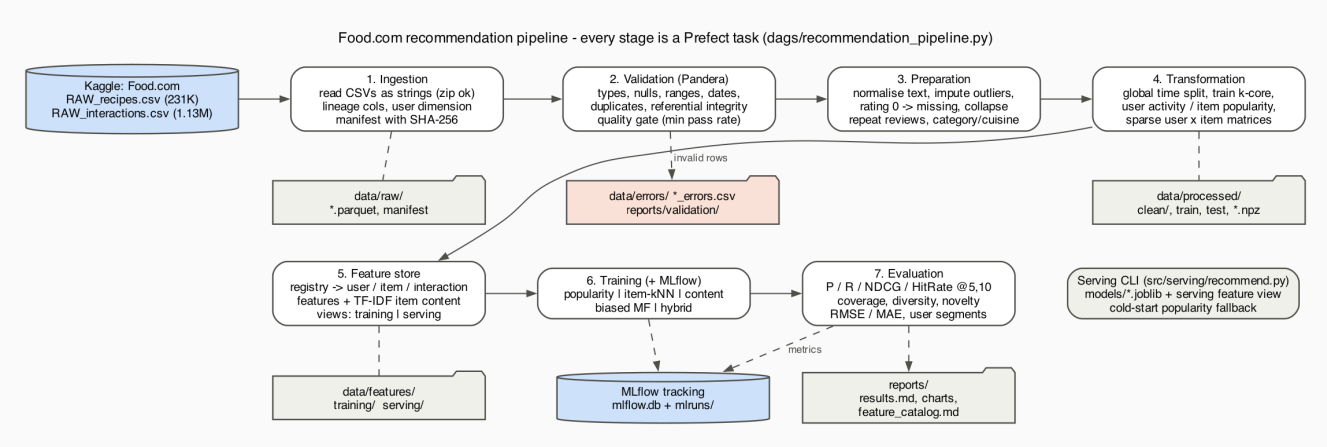
End-to-End Data Management Pipeline for a Recipe Recommender

DM4ML Assignment 2 · Yash Sharma · Roll No. 2025AG05572 · Repository: _____

1. Problem and data

I built a recommendation pipeline for a recipe website: given what a user has reviewed so far, the system ranks recipes they are likely to review (and rate highly) next. I used the public **Food.com** dataset from Kaggle (Majumder et al., 2019): 231,637 recipes (tags, ingredients, nutrition, prep time, steps) and 1,132,367 reviews from 2000–2018 with 0–5 star ratings. Food.com has no user profile table, so the ingestion stage reconstructs a user dimension (236,719 accounts) from everyone who wrote a review or uploaded a recipe. The data is very sparse and very positive: 76% of star ratings are 5, the mean is 4.66, and the median user has only one review.



2. Design decisions and tools

Ingestion. Both CSVs are read straight from the Kaggle zip *as strings*, so badly typed values get caught by validation instead of crashing the reader or being silently coerced. Every row gets lineage columns (`_row_id`, `_source_file`), and a manifest stores the SHA-256 of each source file, so any run can be traced back to the exact input. The data is stored as Parquet in `data/raw/`.

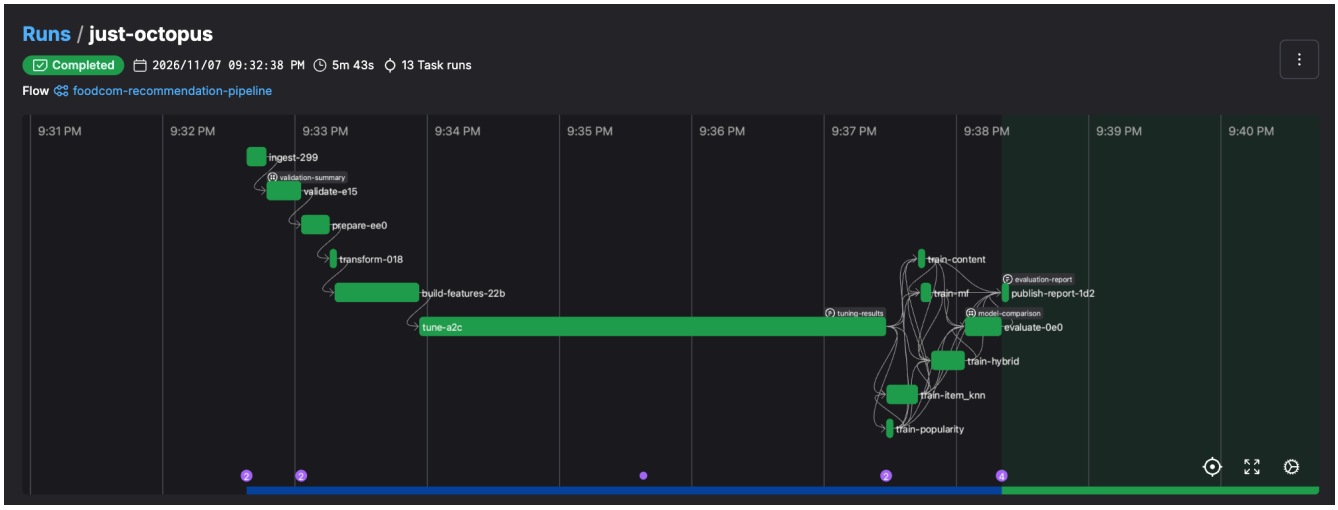
Validation (Pandera + custom checks). Each table is first type-coerced column by column (type errors are recorded), then checked against a Pandera schema (nulls, positive ids, allowed ratings 0–5, a date window of 1999–2018, valid list literals, 7-value nutrition), and finally passed through custom checks: exact duplicates, duplicate keys, referential integrity (review → recipe, review → user) and timestamp consistency. 72 rules are checked in total. Rows that break an *error* rule are written to `data/errors/` with the list of rules they broke. *Warnings* keep the row, and preparation handles them. A quality gate stops the Prefect flow if any table's pass rate falls below 95%. On the real data, the only error rule that fired was an invalid timestamp: **1,246 reviews are dated before their recipe was submitted**, and these were quarantined. Warnings included 4,979 missing descriptions, 1,094 recipes with 0 minutes, 70 with more than 30 days (one says 2,147,483,647 minutes), 125 with more than 10,000 calories, and 60,847 reviews without stars.

Preparation and transformation. Text and tags are normalised. Implausible minutes and calories are set to missing and imputed with the category median (1,164 and 125 recipes). A primary category and cuisine are derived from tags. Rating 0 means "review without stars", so I store it as a *missing rating* but keep the review as an implicit signal. I used a **global time-based split**: the latest 10% of reviews (from 26-Feb-2014) form the test set. The 5-core filter is applied to train only, giving 525,688 reviews from 16,865 users on 39,405 recipes (density 0.079%). Evaluation is warm-start: 2,235 test users with 11,453 relevant (≥4 stars) reviews. Derived datasets are a user activity summary, item popularity, and sparse user×item interaction, rating and review-date matrices, all computed from train only.

Feature management. Instead of Feast I wrote a small feature store. `registry.py` defines 36 documented features: 15 user features (activity count, average rating, days since last review, preferred category, category entropy, …), 15 item features (popularity, Bayesian rating, recent popularity, age, category, minutes, a TF-IDF content vector, …) and 6 interaction features. The same `compute_features(as_of)` code materialises a **training view** (as of the cutoff, so it is point-in-time correct) and a **serving view** (all data). `FeatureStore` is the only read path for training, evaluation and the `recommend.py` CLI, which falls back to recent popularity for unknown users. The catalog in `reports/feature_catalog.md` is generated from the registry.

Models and tracking. (1) A **popularity** baseline, using review counts from the feature store. (2) **Item-kNN** collaborative filtering: cosine similarity with shrinkage and top-N neighbours. (3) **Content-based**: TF-IDF over tags and ingredients. (4) **Biased matrix factorisation**: mini-batch SGD on star ratings. (5) A **hybrid** that blends item-kNN, content and popularity, using the `user_activity_count` and `user_preferred_category` features. Hyperparameters are chosen by a **tuning stage that never touches the test set**. It carves a validation window out of the end of the training period, rebuilds features as of that inner cutoff, and logs 35 trials as nested MLflow runs. Every model run logs its parameters, data statistics, fit time, MF learning curve, model file, config and feature metadata to **MLflow**. Evaluation then logs its metrics back to the same run.

Orchestration and reproducibility. `dags/recommendation_pipeline.py` is a **Prefect** flow with 13 tasks: ingest → validate → prepare → transform → features → tune → 5 × train → evaluate → publish. It has retries on ingestion, the quality gate, and table and markdown artifacts in the UI. `--serve` registers a weekly schedule. A full run takes about 5 min 45 s on a laptop (screenshot below). All settings live in one YAML file, and the run is reproducible through fixed seeds, pinned `requirements.txt`, source checksums, and a pipeline run id stored on every MLflow run. `make sample` runs the same code on a small synthetic sample in the same file format, and pytest covers the metrics, the validation rules, the quality gate and the absence of leakage across the split.



3. Results

Test window 26-Feb-2014 to 20-Dec-2018. Seen recipes are excluded. Coverage is the share of the 39,405 train recipes that appear in any top-10 list. Diversity is 1 – mean pairwise TF-IDF cosine similarity within a list.

Model	P@10	R@10	NDCG@10	HitRate@10	NDCG@5	Coverage@10	Diversity@10	RMSE	MAE
Popularity (baseline)	**0.0068**	**0.0246**	0.0148	**0.0591**	0.0110	0.0011	0.913	0.629	0.420
Item-kNN	0.0063	0.0231	0.0152	0.0559	0.0122	0.0233	0.908	–	–
Content (TF-IDF)	0.0006	0.0020	0.0013	0.0058	0.0009	**0.1028**	0.649	–	–
Biased MF	0.0001	0.0003	0.0004	0.0009	0.0004	0.0073	**0.953**	**0.587**	**0.358**
Hybrid	0.0066	0.0214	**0.0164**	0.0564	**0.0138**	0.0023	0.902	–	–

Predicting the global mean rating gives RMSE 0.639 and MAE 0.437.

- **Popularity is a very strong baseline here.** It has the best precision, recall and hit rate. The hybrid wins on NDCG@10 (+11%) and NDCG@5 (+25%), so it puts the hits higher in the list, but it does so with little coverage, because tuning gave popularity 60% of the weight.
- **Item-kNN is the most balanced personalised model.** It matches popularity on NDCG while covering 21× more of the catalog. Tuning made a big difference: down-weighting very heavy reviewers (some have more than 2,000 reviews) improved validation NDCG@10 by about 60%. Item-kNN works best for users with 10–49 reviews (NDCG@10 0.019 vs 0.016 for popularity), while popularity stays better for users with 5–9 reviews.
- **Matrix factorisation is good at rating prediction but useless for ranking.** It has the lowest RMSE (8% below the global mean), but because almost every rating is 4–5 stars, its item biases mostly pick obscure recipes with a couple of 5-star reviews. This is the well-known gap between RMSE and top-N accuracy. Content-based filtering has the highest coverage but low accuracy, because many tags are generic (e.g. "60-minutes-or-less").

4. Limitations and future work

- **Absolute scores are low.** NDCG@10 ≈ 0.016, because the matrix is 99.92% empty and the test window spans almost five years of taste and catalog drift. **79% of test reviews come from users who were new, or had fewer than 5 reviews, before the cutoff**, so they are excluded from the warm-start metrics. In the CLI those users get the popularity fallback, but its quality is not measured.
- **Evaluation design.** I used one global split and report no confidence intervals or significance tests, and the hyperparameter grid is small. A rolling-window split and bootstrapped intervals would make the model comparison more trustworthy.
- **Ratings and relevance are close to "reviewed".** Ratings are so positive that "relevant = 4+ stars" means almost any review. Treating star-less (0) reviews as missing is my own interpretation. An implicit-feedback model (ALS or BPR) would probably rank better than explicit MF.
- **User data is thin.** The reconstructed user table has no demographics, so user features come only from behaviour.
- **Infrastructure is local only.** The feature store is offline Parquet with full recomputation (no online store or incremental updates), MLflow and Prefect run locally, and data versioning relies on checksums rather than a tool like DVC. Adding Feast or another online store, DVC, and a scheduled Prefect deployment with drift monitoring would be the next steps.